Shiva Shukla

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SUMMARY

Software Engineer with 6+ years of experience building scalable, secure backend systems in Java, Spring Boot, Spring WebFlux, Python FastAPI, SQL/NoSQL databases, and accessible front-end components using React, Lit, and TypeScript. Experienced in MCP Servers, AI Agents, LLMs, microservices, REST APIs, OAuth2/JWT security, Redis cache, OpenTelemetry observability, Swagger, automated testing, and CI/CD through GitHub Actions, GitLab, Jenkins, Helm Charts, Terraform, and Azure Kubernetes in Agile.

WORK EXPERIENCE

PTC Inc. **Feb 2026 - Oct 2026**

Software Engineer Specialist (Full-Stack with Python, Java, TypeScript, Lit, Azure, SQL) *Wayne, PA*

- Built an Azure-based **FastMCP** gateway that exposes registered agent tools via **JSON-RPC 2.0** and **REST APIs**, with **OAuth2/JWT**, request validation, and standardized error handling. Added **Redis** and in-memory caching for an **Azure Storage** registry to improve agent discovery and routing.
- Instrumented FastAPI services with **OpenTelemetry** (traces, metrics, logs) exporting to Datadog and Prometheus for **SLA/SLO** alerting, with trace-log correlation, a circuit breaker on the **async export queue**, and a telemetry health monitoring REST API.
- Automated **CI/CD** pipelines using **GitHub Actions** and **GitLab** with reusable components, integrating **HashiCorp Vault**, **Wiz**, **Black Duck**, and **Checkmarx**, and deploying services on **Azure Kubernetes** using **Helm charts** and pre- and post-deployment tests.
- Provisioned **Azure infrastructure** (AKS, Storage, Redis) as code with reusable **Terraform** modules in **GitLab**, cutting environment setup time by 20% and eliminating config drift across environments.
- Secured the Requirement Assistant (RA) AI tool plugin with **Spring Security** (**OAuth2/JWT**) and token rotation policies, reducing unauthorized requests by 15%.
- Improved test coverage for the RA tool from 30% to 80% with **Playwright**, **JUnit**, **Pytest**, and containerized integration tests with **GitHub Copilot**, **Claude**, and LLM-driven **prompt engineering**.
- Obfuscated 10+ core LLM algorithms using **Cython** to secure intellectual property for a **private SaaS** deployment, keeping latency within 3% of the Python baseline.

PTC Inc. **Aug 2023 - Jan 2026**

Associate Software Engineer (Full-Stack with Python, Java, TypeScript, Lit, Azure, SQL) *Wayne, PA*

- Designed and developed **localized**, **WAI-ARIA-compliant TypeScript** and **Lit** web components for the RA AI tool in a software project management application, increasing accessibility by 50%.
- Developed the RA AI tool plugin using **Java**, **OOP**, **Spring Boot**, **REST APIs**, **Gradle**, **OAuth2/JWT**, **Tomcat**, and **Postgres** to mediate communication between the web component and backend LLM services, improving request processing throughput by 25%.
- Co-developed a **FastAPI** application using **LangChain** and **OpenAI LLMs** to evaluate/rewrite software requirements as per **INCOSE** standards, reducing token costs by 25% and improving F1 score by 22% through **prompt optimization**.
- Integrated an API in the RA plugin to send API metadata to **Snowflake** for faster failure triage.
- Optimized **stored procedures** in a **MySQL** database, improving the GET API performance by 30%.

GlobalLogic **Jul 2018 - Aug 2021**

Software Engineer (Backend Engineer with Java, Spring Boot/WebFlux, NoSQL) *Noida, India*

- Developed **REST APIs** for a learning platform using **Java**, **Spring Boot**, **Reactive Streams**, **Kafka**, and the **Couchbase NoSQL** database; added unit tests with **JUnit**, **Mockito**, and **StepVerifier**, improving code quality by 35% and test coverage by 40%.
- Implemented **Kafka** with **Spring Cloud Streams API** and a retry mechanism to improve **microservice** reliability by 30%.
- Optimized **Couchbase N1QL** queries with **composite indexes**, improving lookup by 25% across 5 high-traffic REST endpoints.
- Authored consumer contracts using Spring Cloud Contracts, reducing early requirement gaps by 20%.

SKILLS

- **Languages**: Java, Python, JavaScript, TypeScript, SQL, HTML5, CSS, React
- **Cloud**: Azure Kubernetes Service, Azure Storage, AWS – S3, EC2, RDS, DynamoDB, Lambda, CloudWatch
- **Frameworks**: Hibernate, Spring Boot, Spring Cloud, JUnit, Mockito, TestNG, ShellSpec, Lit, Playwright, FastAPI
- **Security**: OAuth2, JWT, Spring Security, HashiCorp Vault, Wiz, BlackDuck, Checkmarx, Invicti, SonarQube
- **Databases**: Couchbase, MySQL, PostgreSQL, Redis
- **AI**: Model Context Protocol (MCP), FastMCP, AI Agents, LangChain, LLM, Prompt Engineering, Claude, GitHub Copilot
- **Tools and Technologies**: Gitlab, Jenkins, Git, Docker, Kubernetes, Apache Tomcat, Kafka, Zookeeper, Maven, Gradle, Agile, Linux, Microservices, Helm, Terraform, OpenTelemetry, Prometheus, Datadog, Snowflake, Restful API, Swagger, GitHub Actions

EDUCATION

George Mason University **Aug 2021 - May 2023**

Master of Science, Computer Science (GPA - 3.93) *Fairfax, VA*

Harcourt Butler Technological Institute **Aug 2014 - Jun 2018**

Bachelor of Technology, Information Technology (GPA - 3.8) *Kanpur, India*